



GURU D

Full-Stack Engineer — Distributed Systems • Security • Applied NLP

Vellore, TN • +91 80727 79893 • dguru5079@gmail.com
linkedin.com • github.com

SUMMARY

Built production-grade distributed systems and zero-knowledge security architectures. Seeking SDE/Mobile/Backend roles to ship high-impact features.

SKILLS

Languages

TypeScript, Java, Python, C, C++

Backend

Node.js, Express.js, Redis, Socket.io, REST APIs

Frontend

React.js, Flutter (Dart), Vite, Tailwind CSS

Databases

MongoDB, PostgreSQL (Supabase), SQLite, Firebase

AI/ML

NLP, Sentence-BERT, Semantic Similarity, Siamese Networks

Security

AES-256, Zero-Knowledge Architecture, JWT, Penetration Testing

Tools

Git, Docker, k6, Playwright, Vitest, Jest, Google Gemini API

PROJECTS

Zoomcab

Distributed Ride-Hailing Platform

Node.js • React • Redis • Socket.io • Google Gemini API

- Architected real-time ride-matching engine using Redis Geospatial ($O(\log N)$)—eliminated third-party matching dependency
- Eliminated race conditions with atomic Lua scripts for check-and-set driver assignment
- Implemented Bully Algorithm for leader election with <10s failover detection for surge pricing coordination
- Integrated Gemini API for AI concierge; applied Haversine formula for geodesic fare accuracy

Zero-Vault

Zero-Knowledge Password Manager

React 19 • TypeScript • Node.js • Supabase • Native C11

- Engineered zero-knowledge architecture: master password never leaves client; AES-256 encryption/decryption happens pre-transmission
- Wrote native C11 risk engine with immutable hash-chain verification for tamper-evident audit logging
- Built 5-layer conflict resolution (LWW merge, encrypted tombstones, versioned sync) ensuring eventual consistency across devices
- Testing: C unit tests (memory protection), Jest integration, Playwright E2E, k6 load testing

HealNet

AI-Integrated Social Health Platform

MERN • Sentence-BERT • Siamese Network • MongoDB

- Architected AI core using Sentence-BERT in Siamese Network to match patient symptoms against case library via cosine similarity
- Built admin moderation pipeline with structured content review and MongoDB audit logging
- Developed Treatment & Hospital Comparison module analyzing success rates, risk factors, and recovery metrics

EDUCATION

B.Tech Computer Science Engineering

Jul 2023 – Jul 2027

Amrita Vishwa Vidyapeetham, Coimbatore

CGPA: 6.44/10

Relevant Coursework: DSA, DBMS, Operating Systems, Distributed Systems, Cryptography, Full Stack Frameworks

CERTIFICATIONS

- Ultimate AWS Bootcamp — School of DevOps
- Complete Flutter & Dart Development — Udemy